

Course Code : MCS-230
Course Title : Digital Image Processing and Computer Vision
Assignment Number : MCAOL(IV)/230/Assign/2025-26
Maximum Marks : 100
Weightage : 30%
Last Dates for Submission : 31st October, 2025 (For July, 2025 Session)
15th April, 2026 (For January, 2026 Session)

This assignment has sixteen questions of 5 Marks each; answer all questions. Rest 20 marks are for viva voce. You may use illustrations and diagrams to enhance the explanations. Please go through the guidelines regarding assignments given in the Programme Guide for the format of the presentation.

Q1: Explain the Optical, Analog, and Digital Image processing. **(5 Marks)**

Q2: (i) What should be the physical size of a 2D image of a document with dimensions is 3200x2400, when scanned at 400dpi. Here dpi stands for dots per inch. **(2x2.5=5 Marks)**

(ii) If the physical size of a medical image is 4×4 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better-quality image? Will an image of size 512×512 be enough?

Q3: (i) What is the storage requirement for a 2048 x 2048, 24-bit color image? **(2+3=5 Marks)**

(ii) Explain Intensity, Contrast, Brightness, Noise and Resolution with respect to Images.

Q4: (i) Check whether the matrix $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$ is unary or not? **(2.5+2.5=Marks)**

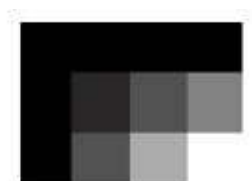
(ii) Perform the transformation $g_1(v) = 3v$ on the image $f(x,y) = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

Q5: Apply the given 3x3 mask w on the following image F (x, y) to generate the new image.

(5 Marks)

$$F(x, y) = \begin{bmatrix} 10 & 3 & 1 & 2 & 20 \\ 2 & 4 & 7 & 2 & 2 \\ 5 & 2 & 2 & 4 & 6 \\ 1 & 4 & 5 & 3 & 7 \\ 6 & 1 & 6 & 5 & 8 \end{bmatrix}, w(i, j) = \frac{1}{9} * \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q6: What is a histogram? Find the histogram of the image given below: **(5 Marks)**



0	0	0	0
0	1	2	3
0	2	4	6

- Q7:** Why is DCT important for image compression? Explain with the help of an example. **(5 Marks)**
- Q8:** What do you understand by Image enhancement? Explain the techniques of image enhancement with a suitable example. Also, discuss the advantages of image enhancement. **(5 Marks)**
- Q9:** Explain the following Smoothing Filter(s): **(5 Marks)**
- (i) Ideal Low Pass Filters (ILPF) (ii) Butterworth Low Pass Filters (BLPF)
- (iii) Gaussian Low Pass filters (GLPF)
- Q10:** What do you understand by feature extraction? What are its applications? Also discuss a few traditional methods of feature extraction. **(5 Marks)**
- Q11:** Explain image degradation and its types. **(5 Marks)**
- Q12:** Transform the RGB cube into its CMY cube. Label all the vertices. Also, interpret the colours at the edges with respect to saturation. **(5 Marks)**
- Q13:** Do you mean by Camera Calibration? Explain how intrinsic and extrinsic parameters of a cameras are estimated? **(5 Marks)**
- Q14:** Explain Bayesian Classification with the help of a suitable example. **(5 Marks)**
- Q15:** Explain K-means clustering methods with the help of a suitable example. Also, discuss the advantages and disadvantages of k -K-means clustering methods. **(5 Marks)**
- Q16:** Perform partitional clustering using Froggy's method for the data given in the table below with k-2 (two clusters). Use first two sample points (3,3) and (6,8) as seed points. **(5 Marks)**

S. No.	X	Y
1	3	3
2	6	7
3	8	10
4	4	5
5	6	6
6	12	10
7	15	14
8	18	16

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MCS-231 Mobile Computing MCAOL (IV)/231/Assign/2025-26

Q. Explain Thick client Architecture and Thin Client Architecture, Also explain the difference b/w them.

Ans:- Thick Client Architecture:

it is known as fat client is a type of computer that performs the majority of processing and computation tasks locally on the device itself. rather than relying heavily on a server. In this architecture the client machine (such as a desktop or mobile device) houses most of the s/w components and resources necessary for running applⁿ.

Characteristics:-

1. Offline Capability:- One of advantages of Thick Clients is that They can often operate without a continuous internet connection. Since most of the data and processing power is local, users can still use applications and access resources even when disconnected from the server. This is useful for mobile computing environments where users may not always have access to high-speed internet.

2. Local Processing:-

The client handles significant portion of data processing, business logic and application execution. While the server may still be involved in data storage or large-scale operation, the client is responsible for the heavy lifting.

3. Rich User Interface:-

Thick clients generally have richer, more interactive user interfaces compared to thin clients. This is because They can leverage the client's local processing power and resources.



Examples:-

• Ms Office Suite:-

Programs like word, excel, or powerpoint are examples of Thick client applications.

- Video Games:-

Many high-performance video games, when substantial rendering and processing is done locally, also function on Thick client architecture.

- Thin Client Architecture:

a Thin client architecture refers to a setup in which the client device is minimalist and relies heavily on a Server for most of its processing and data storage.

The client typically only handles basic user interface operations such as displaying data and receiving user input.

The Server performs most of the computation and manages the application.

Characteristics:-

2: Dependence On Server:-

Since the majority of computing is done on the Server, Thin Clients require constant connectivity to the Server to function effectively. If the Server is down or the network is unavailable, the Thin Client may not be able to provide any services to the users.

Minimal local Processing:-

Thin clients are designed to do minimal local processing. The user device merely sends input data (like keystrokes and mouse movements) to the Server and the Server responds by rendering the output which is sent to the client.



3- low Hardware Requirements:-

The client devices used in a thin client architecture are typically low-powered and less expensive. They don't need high performance processors or large amounts of RAM because they are just rendering the server output and sending inputs.

4- No offline capability:-

Since most of the processing occurs on the server, thin clients are generally unable to function when disconnected from the network.

Examples:-

- web Browsers:-

Most web applications that run in a browser are thin client applications. All of the heavy lifting happens on the server, and the browser simply displays the result.

- Virtual Desktop Infrastructure.

Solutions like Citrix or VMware where the user accesses a virtual desktop that runs on a server rather than on the local machine.

— Differences b/w Thick Client & Thin Client.

- Thick Client

- Handles most processing locally.
- High- requires powerful devices.
- can function offline.
- More difficult to manage.
- Typically higher, due to powerful devices.
- Richer, more interactive UI.
- Less scalable.
- Potentially more vulnerable to local breaches.



- Thin Client:-
- Relies on the Server for processing
- Low- require minimal hardware.
- Requires constant connection to the Server.
- Easier to manage.
- Lower cost, as the Client device is simpler.
- Simpler UI with minimal features.
- Highly scalable with minimal client hardware.
- Centralized Security and updates.

- Q2: What are Smart Sensors? what are Actuators?
explain the differences b/w them.

- Answer:-

In today's world of mobile computing, automation, and the Internet of Things (IoT) Smart Sensors, and actuators play a crucial role in how machines interact with the physical world. Whether it's your Smartphone adjusting its screen brightness based on ambient light, or a smart home system turning on the heater when the temperature drops, smart sensors and actuators are silently working behind the scenes to make it all possible.

- What are Smart Sensors:

is an advanced device that not only detects or measures physical input from the environment (like temperature, light, motion, etc.) but also processes that information and sends it to a connected system or device. Unlike traditional sensors, smart sensors include built-in electronics that can filter, amplify, and interpret the raw data before transmitting it, usually through wireless communication.



Components of a Smart Sensors.

1. Sensing Element:

The Core Component that detects changes in the environment
for example :- a photodiode in a light sensor.

2. Analog to digital Converter (ADC)

Converts the sensor raw data (often in analog form) into digital signals.

3. Processing Unit:

This unit processes and analyzes the data locally. It can filter out noise, apply algorithms, and even make decisions.

4. Communication Interface:

Allows the sensor to send the processed data to other devices such as a mobile app, cloud system, or embedded controller via Wi-Fi, Bluetooth, Zigbee, or another protocol.

Example of Smart Sensors:

- Accelerometer - in smart phone that detects orientation
- Heart rate monitor - in a smartwatch
- Proximity Sensor: that turns off your phone screen during a call.
- Temperature and humidity sensors in a smart thermostat.

— what are actuators?

While sensors are the "eyes and ears" of a system, actuators are the "hands and feet." An actuator is a device that converts an electrical signal into a physical action - such as movement, force, or control of a mechanism or system. It receives commands from a controller or processor (which might be receiving data from sensors) and takes action based on those inputs.



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1. **Electrical Actuators**: use electric current to create motion
2. **Hydraulic Actuators**:
use fluid pressure to move or control a mechanism.
3. **pneumatic Actuators**:
Use compressed air for movement.
4. **Thermal & Magnetic Actuators**:
Use heat & magnetic force for actuation.
Examples:-
 - **Vibration motor**:- in a mobile phone that triggers haptic feedback.
 - **Servo motors**:- in robotic arms that move based on instructions
 - **Valve actuators**:- in smart irrigation system that open or close water flow.
 - **Speakers**:- which convert electric signal to sound.

— **Difference b/w Smart Sensor and actuators:-**

- **Smart Sensor:-**

- Detects and measure physical inputs
- Collects data from environment.
- Often includes onboard processing and logic.
- Accelerometer detects phone movement.
- Monitor and sense conditions.
- Operates independently as part of a system.
- Typically low power.

— **Actuator:-**

- Executes physical actions in response to commands.
- Performs action based on control signal.
- Usually responds to control signals, minimal processing
- Vibration motor gives tactile feedback.
- Act or move based on conditions or instructions.



Q3. What is need for Mobile IP? Explain the working of mobile IP

Ans:- In today's mobile world, Smartphones, laptops, tablets and other portable devices have become essential for communication work, and entertainment. One of the key features that makes mobile devices so versatile is their ability to connect to different networks on the go.

Whether it's switching b/w wifi hotspots, moving from a cellular network to a private network, or roaming across different geographic regions. However, this seamless mobility presents a challenge when it comes to maintaining continuous, uninterrupted communication as we move b/w networks.

- Mobile IP is a protocol developed by the Internet Engineering Task force to allow mobile devices to maintain their IP address and stay connected to the internet even as they move b/w different networks.
 - How does Mobile IP work -
Mobile IP works by enabling a mechanism where communication b/w the mobile device and the rest of the internet doesn't get interrupted when it changes networks. This is achieved through a combination of the Home agent and Foreign agent, along with a process known as Tunneling.
- Components of Mobile IP:
1. Home Agent:- A router on the mobile device home network responsible for keeping track of device location.
 2. Foreign Agent:- A router on the network the mobile device is visiting. It provides a temporary care-of-Address to the device it is visiting.
 3. Mobile Node:- The device that is moving b/w networks. It uses its Home address for communication, but when it moves to a different network.



Mobile IP Process:-

1. Registration:- when a mobile device moves from its home network to a foreign network, it sends a registration request to the home agent. The home agent then maps the mobile device's home address to the newly assigned care-of address in the foreign network.

2. Tunneling:-

The home agent forwards any incoming data packets destined for the mobile device's home address to the device's current care-of address. This is done through a process called tunneling, where the data packets are encapsulated and sent to the foreign agent or directly to the device, depending on the step.

3. Data Forwarding:-

The foreign agent upon receiving the tunneled data packets, delivers them to the mobile device.

4. De-registration:-

When the mobile device moves again, it sends a de-registration request to its home agent and updates its location by registering a new care-of address.

Mobile IP Operations:-

- Seamless Communication:

Mobile IP allows ongoing communication (like VoIP calls) to continue uninterrupted as the mobile device moves between different networks.

- Low Latency:-

Even though there is a slight delay in data forwarding due to tunneling.

Support for Roaming:-

Mobile IP enables devices to roam across different networks.



Q. what is Adaptive clustering in mobile Computing:-

- In mobile computing, the dynamic nature of wireless networks characterize by mobility, varying node density, and fluctuating signal quality. presents significant challenges for maintaining efficient communication. One of the techniques used to tackle these challenges is adaptive clustering, but what is it and how does it help improve the performance.

At its core, adaptive clustering is a technique used in wireless networks (such as mobile ad hoc networks, sensor networks, and large scale IoT systems) to group devices (nodes) into clusters that can work together to optimize network resources, manage communication and reduce energy consumption.

- why is Adaptive clustering

The demand for wireless network is growing rapidly due to the increase in mobile devices IoT systems, and wireless sensors. where each device communicates directly with all others, are inefficient in their dynamic, mobile environment due to several reasons.

1. Network Congestion:

when all devices try to send data to one another directly the network becomes congested leading to packet loss, delays and inefficient use of bandwidth.

2. Energy Efficiency:-

Mobile devices typically have limited battery life. Direct communication between all devices can drain power quickly, requiring strategies to reduce power consumption.

3. Scalability:-

As more devices join a network, maintaining communication efficiency without overloading becomes increasingly difficult.



4. Dynamic Network Condition:

Mobile nodes can move, change their connectivity and experience fluctuation in signal quality.

Static networks are unable to adapt to these changes in real-time.

How does Adaptive clustering work?

The process of adaptive clustering solution can be broken down into two main components:

1- Cluster Formation:

In a mobile wireless network, nodes are initially scattered across a geographical area.

Clustering algorithms are used to form groups.

or clusters based on various criteria like proximity, communication range, or energy levels. These clusters enable efficient communication between devices.

→ Cluster Head (CH)

Each cluster typically has one cluster head, which acts as the central point of contact for the nodes within. The cluster head is responsible for managing communication, aggregating data, and forwarding information to other clusters or the central server.

→ Member nodes:-

The remaining nodes in the cluster are called member nodes. They communicate with the cluster head and follow the management and control decisions made by the cluster head.

• If nodes move closer together or further apart, the clustering algorithm may reassign members to new cluster heads or reorganize the network.

• The algorithm may consider factors such as signal strength, battery power, and network load to determine the most optimal cluster configuration.



2. Cluster Maintenance:-

Once clusters are formed, maintaining the cluster becomes crucial for ensuring the network remains functional and efficient as nodes move or change their communication status. Adaptive clustering mechanism regularly monitors network condition and dynamically update the clusters as needed.

- Cluster Reorganization:-

If a node moves out of range of its current cluster the system will reorganize the network by creating new cluster or merging existing ones to maintain effective communication.

- Cluster head Rotation:-

Since the cluster head has a significant responsibility in terms of data aggregation and communication, it's often beneficial to rotate the cluster head role among nodes to balance the workload and extend the network overall battery life.

- Link Quality Monitoring:-

The system continuously monitors the link quality b/w nodes and adjusts the cluster to avoid weak links that may cause delays or packet loss. If a cluster head or a member node moves out of a region with good coverage.

- Benefits of Adaptive clustering:-

1. Energy efficiency:-

By reducing the need of for long-range communication b/w nodes adaptive clustering helps conserve energy.

2. Improved Scalability:-

Adaptive clustering helps to scale network efficiently. As more devices join the network the clustering algorithm can dynamically adjust.

3. Robustness to node mobility.

4. Load balancing

5. Reduced Communication Overhead.

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